

STRATEGIC IMPLEMENTATION ROADMAP

National Security Operations Center 2026-2030

**Phased Approach to Building World-Class Cyber Defense
Capabilities**

PHASE 1: FOUNDATION (Year 1 - 2026)

The Foundation phase establishes the core infrastructure, organizational framework, and baseline capabilities upon which subsequent phases will build. This phase focuses on creating the physical and logical foundation for SOC operations, deploying essential security technologies, recruiting and training initial personnel, and establishing operational processes and procedures. By the end of Phase 1, the SOC will provide basic 24/7 monitoring coverage for priority government networks and critical infrastructure entities.

1.1 Infrastructure Development

The physical SOC facility will be established in a secure location meeting stringent requirements for physical access control, environmental controls, power redundancy, and network connectivity. The facility design incorporates a main operations floor with analyst workstations, incident response war rooms, executive briefing areas, and secure spaces for handling classified information. Technical infrastructure includes enterprise-grade server systems, high-performance storage arrays, redundant network connectivity with diverse path routing, and comprehensive backup power systems.

Component	Specifications	Quantity	Priority
SOC Facility	500+ sqm, Tier III datacenter standards	1	Critical
Analyst Workstations	Quad-monitor setups, high-performance	25	Critical
Server Infrastructure	Distributed computing cluster	15 nodes	Critical
Storage Systems	Petabyte-scale, encrypted storage	2 arrays	Critical
Network Equipment	10/40Gbps switching, firewall appliances	Full stack	Critical
Power Systems	UPS + Generator backup, 72hr autonomy	Full redundancy	Critical

1.2 Technology Platform Deployment

Phase 1 technology deployment focuses on establishing the core security platform components that form the technological backbone of SOC operations. The primary platform is an enterprise SIEM (Security Information and Event Management) solution capable of collecting, correlating, and analyzing log data from across monitored environments. Complementary deployments include EDR (Endpoint Detection and Response) agents for endpoint visibility, network traffic analysis tools, vulnerability scanning platforms, and basic SOAR (Security Orchestration, Automation and Response) capabilities for workflow automation.

Technology Category	Solution Type	Purpose	Integration Priority
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SIEM Platform	Enterprise-grade (Splunk/QRadar/Sentinel)	Log correlation, alerting	Primary
EDR Solution	Next-gen endpoint protection	Endpoint telemetry, response	Primary
NTA/NDR	Network traffic analysis	Network visibility, detection	Secondary
Vulnerability Management	Continuous assessment platform	Risk identification	Secondary
Threat Intelligence	TI feeds and platforms	Contextual enrichment	Enhancement
SOAR Platform	Orchestration and automation	Workflow automation	Phase 1 Basic

1.3 Organizational Build-up

Building the SOC team requires a strategic approach to talent acquisition, training, and organizational development. Phase 1 targets recruitment of 27 full-time equivalent (FTE) positions covering core operational functions including SOC management, tiered analyst roles, engineering support, and administrative functions. Recruitment strategy emphasizes attracting experienced professionals from both domestic and international markets while simultaneously developing entry-level talent through structured training programs and partnerships with educational institutions.

Role	Count	Experience Level	Training Required
SOC Director	1	Senior (10+ years)	Leadership program
Shift Supervisors	3	Mid-Senior (7+ years)	Operations management
Tier 1 Analysts (L1)	12	Entry-Mid (1-3 years)	SOC fundamentals certification
Tier 2 Analysts (L2)	6	Mid-level (3-5 years)	Advanced threat analysis
SOC Engineers	3	Mid-Senior (5+ years)	Platform specialization
Admin/Operations	2	Various	Process and tools training
Total Phase 1	27 FTE		

PHASE 2: OPTIMIZATION (Years 2-3, 2027-2028)

The Optimization phase focuses on enhancing detection capabilities, expanding operational scope, integrating advanced analytics, and achieving full operational maturity. During this phase, the SOC transitions from basic monitoring to intelligent, proactive security operations incorporating machine learning-driven analytics, automated response capabilities, and comprehensive threat hunting programs. Staffing expands to support 24/7 operations with complete shift coverage and specialized functions including dedicated threat intelligence and threat hunting teams.

2.1 Advanced Analytics and AI Integration

Phase 2 introduces transformative AI and machine learning capabilities that elevate SOC effectiveness beyond traditional rule-based detection. User and Entity Behavior Analytics (UEBA) solutions establish behavioral baselines for users, systems, and network entities, enabling detection of anomalous activities that indicate compromise, insider threats, or policy violations. Machine learning models trained on historical incident data improve alert prioritization, reducing false positive rates while ensuring genuine threats receive appropriate attention. Natural language processing automates log analysis and correlation tasks that previously required significant manual effort.

Capability	Description	Expected Impact	Implementation Timeline
UEBA Platform	User/entity behavior modeling	Insider threat detection +40%	Q1 2027
ML Alert Triage	Intelligent alert prioritization	False positive reduction -50%	Q2 2027
Automated Enrichment	Contextual data gathering	Analyst efficiency +35%	Q3 2027
Predictive Analytics	Threat forecasting models	Proactive defense +25%	Q1 2028
NLP Log Analysis	Automated log interpretation	Processing speed 3x	Q2 2028

2.2 Full Operational Capability

By the conclusion of Phase 2, the SOC achieves full 24/7 operational capability with complete staffing across all tiers and specialized functions. The organization expands to approximately 40 FTE positions, providing robust shift coverage with overlap for knowledge transfer and handoff continuity. Dedicated teams for threat intelligence, threat hunting, and incident response enable proactive security operations that go beyond reactive monitoring. Comprehensive integration with constituent agencies establishes standardized processes for reporting, escalation, and coordinated response.

2.3 Automation and Orchestration Expansion

SOAR capabilities mature significantly during Phase 2, moving from basic workflow automation to sophisticated playbooks that handle routine response actions autonomously. Integration connectors are developed for all major security and IT systems, enabling orchestrated responses that span multiple tools and teams. Common incident types including malware infections, phishing attempts, policy violations, and unauthorized access attempts are addressed through standardized playbooks that reduce mean time to response while ensuring consistent, documented actions.

PHASE 3: MATURITY (Years 4-5, 2029-2030)

The Maturity phase represents the culmination of the SOC development journey, achieving world-class operational status with predictive capabilities, international integration, and continuous innovation. During this final phase, the SOC evolves from a sophisticated security operations center to a true center of excellence that sets regional standards for cybersecurity operations. Focus areas include predictive security analytics, autonomous response expansion, international partnership development, and establishment of research and development capabilities.

3.1 Predictive Security Capabilities

Predictive security analytics represent the frontier of SOC capability evolution, leveraging advanced machine learning models, threat intelligence fusion, and historical pattern analysis to anticipate attacks before they occur. Rather than solely responding to detected incidents, predictive capabilities enable the SOC to identify precursor indicators, assess likely attack scenarios, and implement preemptive countermeasures. This proactive posture fundamentally transforms the value proposition of security operations from cost center to strategic enabler of business objectives.

3.2 Autonomous Response Evolution

Autonomous response capabilities expand to cover increasingly complex incident scenarios, with human oversight reserved for high-severity or novel situations requiring judgment and decision-making. Machine learning models continuously refine response decisions based on outcomes, improving accuracy and effectiveness over time. The autonomous response framework operates within clearly defined boundaries established by policy, with comprehensive audit logging and explainability features that support accountability and compliance requirements.

3.3 International Cooperation and Leadership

As a mature SOC operation, Algeria assumes a leadership role in regional cybersecurity cooperation, participating actively in international threat intelligence sharing forums and contributing to global cybersecurity capacity building efforts. Bilateral and multilateral partnerships with allied national SOC's enable exchange of best practices, joint exercises, and mutual assistance agreements. The SOC serves as a model and resource for neighboring countries developing their own security operations capabilities, strengthening collective regional cyber resilience.

IMPLEMENTATION SUMMARY

Metric	Phase 1 (2026)	Phase 2 (2027-28)	Phase 3 (2029-30)
Budget	\$8.5M	\$12M	\$10M
Staffing (FTE)	27	40	54
MTTD Target	< 15 min	< 5 min	< 2 min
MTTR Target	< 1 hour	< 15 min	< 10 min
EPS Capacity	50K	500K	2M+
Automation Level	Basic	Advanced	Autonomous
Maturity Level	Initial	Developing	Optimizing